AMERICAN JOURNAL OF PHYSIOLOGY
HEART AND CIRCULATORY PHYSIOLOGY

Am J Physiol Heart Circ Physiol 324: H1–H13, 2023.
First published November 18, 2022; doi:10.1152/ajpheart.00611.2022

REVIEW

Excitation-Contraction Coupling, Electrophysiology, and Arrhythmias

Cardiac function, structural, and electrical remodeling by microgravity exposure

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Abstract

Space medicine is key to the human exploration of outer space and pushes the boundaries of science, technology, and medicine. Because of harsh environmental conditions related to microgravity and other factors and hazards in outer space, astronauts and spaceflight participants face unique health and medical challenges, including those related to the heart. In this review, we summarize the literature regarding the effects of spaceflight on cardiac structure and function. We also provide an in-depth review of the literature regarding the effects of microgravity on cardiac calcium handling. Our review can inform future mechanistic and therapeutic studies and is applicable to other physiological states similar to microgravity such as prolonged horizontal bed rest and immobilization.

arrhythmia; calcium; microgravity; ryanodine receptor; space

INTRODUCTION

Extensive research has shown that microgravity exposure significantly affects the cardiovascular system. Impairment of cardiovascular structure and function is regarded as a major health risk affecting astronauts during spaceflight missions (1). These risks include orthostatic intolerance, decreased exercise capacity, and on-orbit cardiac arrhythmias. Hydrodynamic changes due to blood volume redistribution and abrogation of physiological upright blood pressure gradients induced by microgravity conditions result in alterations in cardiac function and structure.

Despite decades of research, however, a clear mechanistic understanding of microgravity-related alterations in cardiac function and structure remains elusive. This review provides an extensive summary of the associated controversies surrounding these concepts to better define the features and mechanisms of the effects of weightlessness on cardiac function and structure. This understanding will be beneficial not only to explore effective interventions for maintaining astronaut heart health, but also for patients who are bedridden for long periods of time.

MICROGRAVITY-RELATED HEMODYNAMIC ALTERATIONS

Fluid Shifts

Pre- and postspaceflight human echocardiography data have demonstrated that fluid shifts may be a key driver of spaceflight-related cardiac alterations (2). Weightlessness from spaceflight causes fluid shifts leading to central volume expansion and greater preload and cardiac output by the Frank-Starling mechanism. These fluid shifts are hypothesized to occur secondary to upper extremity vasodilation, which can further distend cardiac chambers and stimulate further release of vasodilatory peptides (Fig. 1*A*). Interestingly, the reason behind the initial vasodilation is unknown (2). Nonetheless, weightlessness can lead to chest relaxation and thus reduced intrathoracic pressure, increasing cerebral and intrathoracic blood flow. These changes result in central redistribution of blood volume, activating vasodilatory mechanoreceptors. A summary of findings from several studies regarding cardiac output and stroke volume is shown in Table 1.

Fluid shifts alone, however, cannot completely explain orthostatic intolerance of astronauts on return to earth, as fluid reloading back into normal physiological compartments and physical countermeasures do not significantly improve microgravity-related orthostatic intolerance (17). Participants in a 2-wk study were subjected to head-down tilt bed rest testing and split into a group that performed countermeasure exercises, a group that received a dextran infusion after bed rest, and a group that underwent both interventions (18). Only the group that received both interventions recovered orthostatic tolerance, indicating that volume replacement alone was not sufficient to combat orthostatic intolerance.

Despite the observation of distended jugular veins after microgravity exposure, Buckey et al. (19) demonstrated that

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Submitted 25 October 2022 / Revised 14 November 2022 / Accepted 15 November 2022

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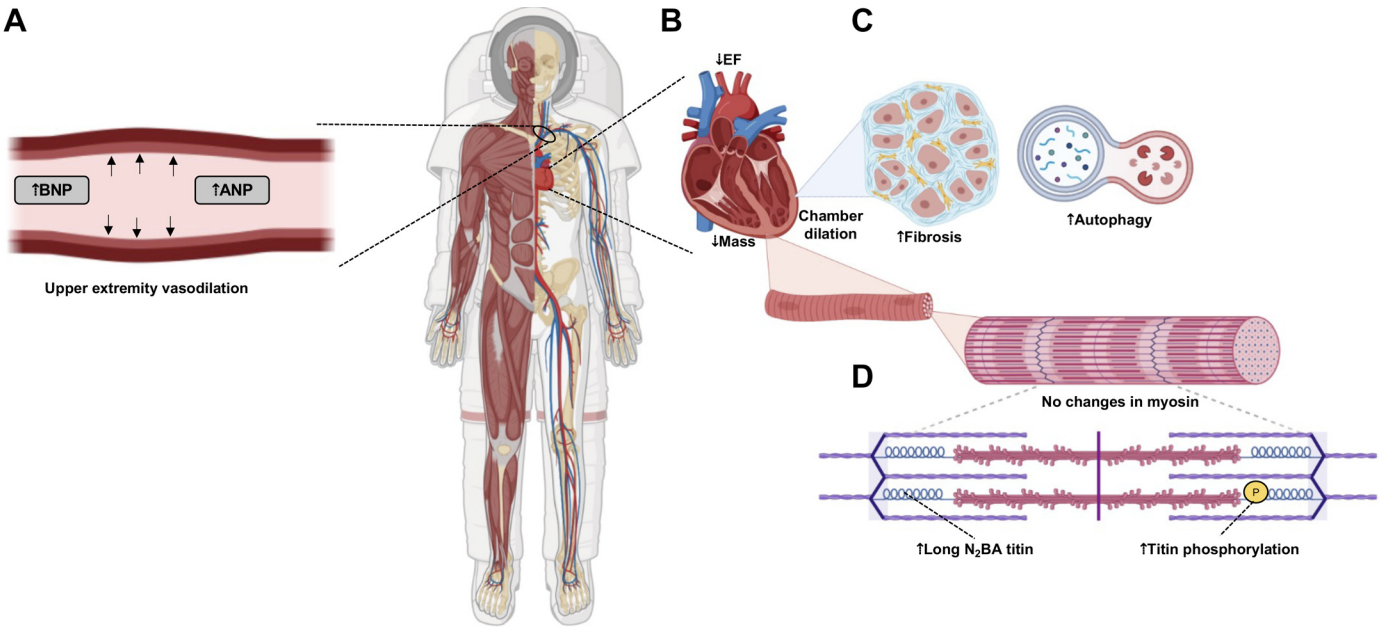

Figure 1. Microgravity-related effects on cardiac structure and function. *A*: microgravity leads to selective upper extremity vasodilation because of release of vasodilatory peptides such as atrial natriuretic peptide (ANP) and B-type natriuretic peptide (BNP). *B*: microgravity results in deceased left ventricular (LV) and right ventricular (RV) ejection fraction (EF) with biventricular dilation and atrophy. *C*: microgravity results in increased fibrosis and autophagy. *D*: microgravity results in no changes in myosin structure but increased long N_2BA titin isoform with greater titin phosphorylation.

central venous pressure is decreased under microgravity conditions via direct intravenous catheter measurements (20). Central venous pressure was directly and continuously measured via catheters in three subjects undergoing spaceflight and decreased within 8 h of spaceflight despite echocardiographic evidence of increased left ventricle (LV) end-diastolic diameter (20). This observation could be secondary to the altered pressure-volume relationship under microgravity conditions, such that the same blood volume can be contained at a lower pressure. Indeed, more recent international space station (ISS) data comparing pre- and post-spaceflight echocardiograms from 12 astronauts similarly point toward the tendency for hearts to take on spherical morphologies (21).

As postflight orthostatic intolerance is a major health concern for astronauts upon returning to earth, studying counteractive mechanisms is clinically important (22). To counteract these changes, Kourtidou-Papadeli et al. (23) suggested using artificial gravity via centrifugal force. They found that greater *g*-load increased vasoconstriction and cardiac output, with male participants having a greater response compared with females (23).

Another recent study of postflight orthostatic challenge, which occurs secondary to blood pooling in lower extremities during long-duration spaceflight, examined the concomitant effects of orthostatic challenge and acute physical fatigue on performance and cerebral oxygenation (22). Sixteen healthy participants performed mental arithmetic tasks and psychomotor tracking with and without a prior 1-h physically fatiguing exercise session. All tasks were performed under orthostatic challenge induced via lower body negative pressure. Cerebral oxygenation increased during mental arithmetic more than psychomotor tracking while under orthostatic challenge. Not surprisingly, the increase in cerebral oxygenation was attenuated in the presence of physical fatigue. However, this effect was more pronounced in male compared with female participants, demonstrating the importance of considering sex in designing countermeasure exercises during spaceflight (22).

Anemia

Anemia, or a decreased red blood cell mass, has been reported to occur after microgravity exposure using ^{51}Cr-tagged red blood cells (24). Within 24 h of microgravity exposure, there can be a 20% decrease in plasma volume, as well as a 10% decrease in red blood cell mass (25). The mechanism is more likely due to an increase in red blood cell destruction, selectively of red blood cells younger than 12 days old (26), rather than a decline in bone marrow production (25). Indeed, a study of 14 astronauts during 6-mo missions aboard the ISS reported greater hemolysis and reticulocytosis that persisted 1 year after landing (27).

MICROGRAVITY-RELATED CHANGES IN CARDIAC STRUCTURE AND FUNCTION

Cardiac Atrophy: Insights from Human Studies

Cardiac muscle is autoregulated in response to changes in loading conditions (28, 29). Greater mechanical loads, such as those seen in obesity (30), exercise training (31), and hypertension (32), lead to increases in cardiac muscle mass. Cardiac chamber volume similarly increases in a load-dependent manner, with highly trained athletes exhibiting increases in LV end-diastolic diameter of ~10%, wall thickness of ~20%, and LV mass of ~45% compared with age-matched controls (33).

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Table 1. *Microgravity-related effects on cardiac function*

Study	Cardiac Output	Stroke Volume	Position	Methodology	Spaceflight Duration
Norsk et al. (2)	↑	↑	Seated	Foreign gas rebreathing technique (before, 85–192 days in space)	3–6 mo
Martin et al. (3)	↓	↓	Left lateral decubitus	Echocardiography 10 days before spaceflight and within 3 h of landing	Short duration (4–17 days), long duration (129–144 days)
White and Blomqvist (4)	↑	↑	Not specified	Three-compartment model of cardiovascular system	Not specified
Peterson et al. (5)	↑	↑	Supine, launch, sitting, standing	Numerical model of the cardiovascular system	Not specified
Pantalos et al. (6)	↓	↓	Upright	UTAH-100 artificial ventricle	40-s periods
Bungo et al. (7)	Cardiac index had no significant change	↓	Left lateral decubitus	Echocardiography (preflight and 1 h after landing)	5–8 days
Mulvagh et al. (8)	No change	↓	Supine, standing	Echocardiography (preflight and 1–2 h after landing)	4–5 days
Gazenko et al. (9)	↓Short term, no change long term	↓Short term, no change long term	At rest on bicycle	Noninvasive tetrapolar rheography	7 days (short term), 65–237 days (long term)
Prisk et al. (10)	↑	↑	Standing	Acetylene foreign gas rebreathing method (preflight and after landing)	9 days
Shykoff et al. (11)	↑	↑	Sitting, supine	Carbon dioxide rebreathing method	9 days or 15 days
Herault et al. (12)	↓	↓	Supine	Echocardiography	5–6 mo
Hamilton et al. (13)	No change	↓	Not specified	Echocardiography (preflight, inflight session)	34–190 days
Hughson et al. (14)	↑	No change	Supine and seated	Finger arterial pressure waves (pre- and postflight)	6 mo
Hughson et al. (15)	↑	↑	Seated	Foreign gas rebreathing method	119–166 days
Marshall-Goebel et al. (16)	↑	↑	Seated	U.S. Doppler	150 days

Cardiac size is reportedly decreased after microgravity exposure (29), but these findings could have several causes, including decreased myocardial tissue mass, chamber volume, and/or alterations in anatomical orientation (Fig. 1*B*). Whether loss of cardiac mass contributes to decreased heart size after spaceflight warrants further investigation. Following the Mercury, Gemini, and Apollo spaceflights, decreased cardiothoracic ratios determined from pre- and postflight X-rays have been reported (34). Nearly 90% (7/8) of crew members exhibited decreases in cardiac silhouette area, but most of them recovered 4 to 5 days after landing. Interestingly, there is no indication that Skylab crewmen, who spent 28–84 days in space, exhibited greater decreases in cardiac size than those observed after prior shorter duration space missions. This observation may have been due to the systematic and intensive countermeasures taken while operating the orbit (34).

Data from four astronauts who did not perform maintenance exercises while in space demonstrated that LV mass, as quantified by cine magnetic resonance imaging (MRI), decreased by 12% after 10 days of spaceflight. Moreover, comparative data from healthy men (mean age, 31 yr) confined to 6 and 12 wk of horizontal bed rest showed that both LV and right ventricle (RV) mass decreased by ~1% per week (Fig. 1*B*). These findings demonstrate that cardiac muscle plasticity under reduced mechanical loading occurs in real life (i.e., spaceflight) and simulated (i.e., bed rest) conditions, and may be a pathophysiological driver of microgravity-related cardiac dysfunction (29).

However, data from long-duration spaceflight missions on the ISS (>6 mo) do not demonstrate that cardiac mass is reduced but rather that cardiac structure seems to be preserved (35). It is important to note that the intense exercise programs (>2 h daily) performed while onboard the ISS seemed to be a major method of ameliorating such detrimental cardiac changes (36). Net heart mass lost during space correlated with the relative mechanical work performed by the heart in space compared with that on earth (37). However, these countermeasures were protective in some, but not all, astronauts (38).

On earth, human head-down tilt bed rest studies have demonstrated that cardiac atrophy occurs at a rate of ~1% per week without countermeasures (3, 39). For example, LV dimensions were significantly smaller in sedentary, intellectually disabled adults compared with their healthy adult counterparts (40). Other studies have used long duration bed rest tests and demonstrated that LV mass was lost at a rate of 1% per week regardless of sex (38). Dorfman et al. (41) simulated maintenance exercise in bed rest subjects on earth via lower body negative pressure. They found no change in LV or RV volumes but did find increased ventricular mass. This could serve as a potential countermeasure for cardiac atrophy.

Decreased cardiac loading, as occurs during spaceflight, can lead to cardiac atrophy (29), which could lead to the changes observed in cardiac size. LV mass directly relates to plasma volume, and microgravity-like cardiac dysfunction

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can be induced by simple dehydration (37). One study restricted salt intake along with other dietary modifications, such as consuming distilled water, in ground-based participants. The results revealed a significant positive correlation between measured LV mass and plasma volume, suggesting that cardiac changes seen in spaceflight, such as decreased LV mass, could be explained by dehydration (37). It is important to note that the use of echocardiography to estimate LV mass in this study is prone to interindividual and interuser variability. Indeed, another study reported different LV mass quantification from MRI when compared with echocardiography (42). Pre- versus postdialysis LV mass was reportedly decreased when quantified by echocardiography, but no changes were detectable when quantified by MRI (42).

Finally, a study of 10 male and 9 female participants examined how cognitive and cardiovascular systems respond to varying gravito-inertial stressors induced via 5-min centrifugation sessions of 2.4 *g* at the feet and 1.5 *g* at the heart (43). Artificial gravity increased heart rate in male participants but decreased heart rate in females. Mean arterial pressure increased in both groups. Participants were also given handheld objects during rotational acceleration to assess grip force modulation. Male participants were better able to modulate grip strength compared with female participants during rotational acceleration (43).

Cardiac Atrophy: Insights from Animal Experiments

Data from rats exposed to microgravity have demonstrated that total body weight remains unchanged whereas heart weight decreases (44). Rats subjected to hindlimb suspension had decreased heart-to-body weight ratios, RV and LV mass, interventricular septal weight, and posterior LV wall thickness. While there was no reported increase in the number of apoptotic cells, greater autophagy was seen in the tail-suspended rats (44).

To evaluate the functional consequences of simulated microgravity on cardiac function, Zhong et al. (45) performed 4 wk of tail suspension hindlimb unloading (HU) in mice followed by 7 or 14 days of recovery. Immediately after HU, whole heart weight was increased likely because of fluid retention despite decreased body weight. LV mass remained unchanged while RV mass decreased immediately after HU and recovered after 7 days.

Importantly, several studies have demonstrated that cardiac cachexia is modulated by biological sex, with postmenopausal females being most susceptible to cachexia secondary to loss of the anti-inflammatory effects of estrogen and lack of testosterone (46). A study of HU in mice reported that female mice had greater muscle catabolism, with concomitantly greater activity of transcription factor forkhead box O3a (FoxO3a) (47). Further supporting evidence of biological sex-driven differences in cardiac size alterations stems from a study of cardiac hypertrophy in mice that demonstrated that regression of hypertrophy depends on the hypertrophic trigger and is modulated by biological sex (48). The triggers studied were angiotensin II and isoproterenol infusion for 7 days. Angiotensin II-treated females did not exhibit regression of pathological hypertrophy following termination of the infusion. On the other hand, isoproterenol-treated males exhibited rapid hypertrophic regression. Pathway analysis demonstrated downregulation of the transforming growth factor b1 pathway in all groups except angiotensin II-treated females following the removal of the hypertrophic trigger (48).

Changes in Cardiac Function

Mice subjected to HU had reduced LV and RV ejection fraction (EF) and fractional shortening despite no changes in LV and RV mass (45). LV EF recovered to normal levels after 14 days of normal gravity conditions following tail suspension. RV EF, however, did not recover after 14 days. Histologic analyses demonstrated evidence of RV and LV fibrosis in the HU mice, with recovery to baseline levels of fibrosis after 14 days. Histone deacetylase 4 (HDAC4) and ERK1/2 phosphorylation were increased after HU; both were restored to baseline phosphorylation levels after 14 days. In the RV, the level of HDAC4 phosphorylation did not restore to normal until 14 days of reloading, whereas this took 7 days in the LV. There was also evidence of autophagy in the HU group (Fig. 1*C*), as demonstrated by the increased LC3-II: LC3-I ratio and decreased AMPK phosphorylation. Neither of these changes recovered after 14 days in the RV. Taken together, these results indicate that the heart undergoes remodeling after exposure to microgravity (Fig. 1*C*). While many of these changes can be reversed, RV changes may be more permanent than those in the LV (45). Further studies are needed to reveal why this is the case.

Another study subjected mice to tail suspension for 28–56 days (49). Echocardiography demonstrated cardiac enlargement and mildly decreased LV EF (57% vs. 65% in the control group) starting at 28 days of HU (Fig. 1*B*). LV EF decreased further to 48% after 56 days of HU, indicative of dilated cardiomyopathy with systolic heart failure (49). In contrast to these findings, Ray et al. (50) reported no short-term effects on cardiac function in rats after microgravity exposure. The study consisted of two arms, the microgravity arm, which divided rats into preflight, flight, flight cage simulation, and vivarium controls, and the HU arm, which divided rats into control, 7 days of HU, and 28 days of HU. Rats in the flight group were exposed to 7 days of microgravity on the Spacelab3 mission. Cardiac function measurements after 28 days HU demonstrated no difference in the maximal rate of rise of the LV pressure ($+dP/dt$) during 5 min of postsuspension standing. Mean arterial pressure and heart rate-pressure product during 5 min of standing were also unchanged, perhaps because of baroreceptor-mediated compensatory increases in heart rate, although the greater sympathetic tone in the HU group could also explain this finding (50). Despite the greater heart rate in the flight group, LV peak $+dP/dt$ was unchanged, indicating impaired LV contractility secondary to dysregulated Ca^{2+} handling and/or decreased preload. After spaceflight, cardiac afterload was also increased because of peripheral vessel remodeling (50).

In contrast to the lack of changes in LV $+dP/dt$ in the HU group from Ray et al. (50), Yu et al. (51) reported reduced papillary muscle contractility following 28 days of HU, indicating differential effects of microgravity on papillary muscle and LV cardiomyocytes. Goldstein et al. (52) reported that rats exposed to microgravity on COS-MOS 2044 had significantly decreased cross-sectional area of myofibers of

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papillary and ventricular muscle samples. However, rats exposed to microgravity had normal A-band and Z-band spacing in myofibrils, suggesting that cardiac tissue architecture was preserved. Other studies, however, did not report changes in LV contractility (8), suggesting a greater effect of microgravity on decreasing preload and/or increasing afterload. These findings are consistent with human studies, which have reported no changes in EF after short-term (8) and long-term (237 days) spaceflight (53) and 10 days of bed rest (54).

In a 4-wk model of tail suspension in rats, contractile response to electrical stimulation was significantly decreased in the HU group compared with controls (55). The inotropic effects of isoproterenol on ventricular myocytes were also decreased in the HU rat group, indicating β-receptor desensitization (55). β-Receptor desensitization also impacted β_2-adrenergic receptors within the vasculature, as evidenced by the attenuated decrease in blood pressure in response to isoproterenol in the HU group (55). There were no changes in heart weight, body weight, or ambient blood pressure. LV pressure and systolic function, assessed by $+dP/dt_{max}$, were decreased. $G_{s\text{-}\alpha}$, the biologically active form of the G_s protein that mediates β-adrenergic signaling (56), was unchanged, but there was decreased adenylyl cyclase production in response to high-dose adenylyl cyclase activator forskolin (55). While the results of this study seemingly contradict the increased β-receptor sensitivity reported in studies of human head-down bed rest (57, 58), it is important to note the longer timeline and lack of psychological stress in this study (55).

Altogether, morphological changes under microgravity conditions do occur, although different cardiac alterations are reported among studies. These differences elucidate the complicated and detailed mechanisms governing the regulation of cardiac remodeling under microgravity conditions. Moreover, environmental factors such as nutrition, hemodynamic and mental stresses, gravity environment, and radiation are hard to standardize across studies and likely contribute to the different alterations reported among studies. Nonetheless, microgravity induces cardiac dysfunction due to decreased preload, contractility, and altered β-adrenergic response.

Ultrastructural Cardiac Changes

A study on cardiac morphological structure changes in rats under simulated microgravity via 8 wk of tail suspension found that heart weight remained unchanged while mean LV volume decreased (59), changes indicative of cardiac atrophy. These alterations were primarily due to decreased cardiomyocyte length, as the cross-sectional area trended lower but was nonsignificant. Interstitial, endothelial cell, and mitochondrial volume density also decreased, perhaps secondary to lower oxygen demand. The lack of differences in heart weight reported in this study was likely due to biological heart weight variation, as well as transient changes in capillary blood volume (59).

Data from 2-wk unmanned spaceflight experiments in rats revealed that microgravity resulted in decreases in average myofiber cross-sectional area, sarcomere longitudinal section, and myocardial myosin ATPase activity in spaceflight compared with control rats (52, 60). There were also increases in lipid droplets and glycogen storage, along with morphological mitochondrial changes (61). These changes included a reduction in mitochondrial number and volume, as well as increased mitochondrial volume density and mitochondria-to-myofibril ratio (52, 61). A-band and Z-band spacing were normal in spaceflight compared with control rats (52).

Ulanova et al. (62) studied cardiac ultrastructural and striated muscle filament protein changes in mice after 30-day spaceflight. LV myocardial samples exposed to microgravity displayed altered sarcomeric structure, blurring of the A- and I-band boundaries, and an increase in interfilament A-band spacing. These changes resemble features of ischemia-reperfusion in isolated hearts and hypoxic preconditioning, a technique used to increase the body's ability to adapt to low oxygen tension conditions such as outer space (63). Mice exposed to spaceflight exhibited no changes in cardiac and skeletal muscle weight. Intact titin (T1) content and titin phosphorylation in cardiac muscle were also unchanged (62). However, there was an increase in titin gene expression and titin T2-proteolytic fragments in cardiac muscle, indicative of titin degradation (62). Altogether, muscle atrophy seems to occur during spaceflight while sarcomeric alterations may be postflight phenomena secondary to landing-related stressors. Nonetheless, it remains unknown whether these sarcomere changes are adaptative or pathological.

Cardiomyocytes derived from human induced pluripotent stem cells (hiPSC-CMs) aid the understanding of microgravity's effects on the human myocardium and provide a bridge from animal models to human tissue. Weichselbaum et al. (64) cultured hiPSC-CMs and cardiac fibroblasts to form cardiac tissue constructs responsive to physiological stimuli. Experiments using hiPSC-CMs show that simulated microgravity also has effects on the cellular level. A group of cardiomyocytes exposed to simulated microgravity (SMG-CM) formed stress fibers and membrane calveolae, which are typical changes in response to mechanical stress. In addition, these cardiomyocytes also had impairment of mitochondrial membrane potential, which could lead to decreased ATP synthesis. Taken together, these findings suggest that contractility would be affected in microgravity conditions (65). However, although there were some structural changes in SMG-CM, no significant changes in the morphology of their sarcomeres were recorded (65, 66).

Changes in Oxidative Stress and Cell Cycle Regulation

One possible molecular pathophysiological driver behind spaceflight-related cardiac atrophy is increased transcription of oxidative stress-related genes (67). A study of mice who underwent spaceflight for 15 days reported increased expression of oxidative stress pathways in ventricular tissue harvested 3 h after landing. *Nox1* was upregulated, and *Nfe212* and *Ptgs2* were downregulated. *Nfe2l2* is a transcriptional regulator of genes responsible for the oxidative stress response and undergoes rapid transcriptional alterations in response to changes in gravitational acceleration (68). Indeed, *Nfe2l2*-deficient mice were protected from spaceflight-induced changes in oxidative stress-related genes (69). *Nox1* is known to catalyze electron transfer to generate superoxide radicals, thus increasing reactive oxygen species (ROS) and oxidative stress

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(70). *Ptgs2* functions as a peroxidase and plays an important role in preventing accumulation of ROS and was reportedly decreased after spaceflight (67).

Cdkn1a, also known as p21, is upregulated after spaceflight and functions to inhibits cell cycle progression at the G_1 phase, thereby promoting apoptosis (71). Interestingly, cardiac hypertrophy gene *Myc* was also upregulated in spaceflight heart samples while proinflammatory mediator *Tnf* was downregulated, indicating that immunosuppression and abnormal cell cycle regulation potentially contribute to spaceflight-related cardiac dysfunction (67).

MICROGRAVITY-RELATED DYSBIOSIS

Several studies have reported on changes in the microbiome following microgravity exposure, although the majority of studies do not seem to suggest that the microbiome changes are clinically pathological (72). Brown et al. (73) sampled 18 Skylab crewmembers before and after spaceflight and reported an increase in oral anaerobic bacteria during flight but no postflight health consequences. The MARS500 study followed six male crewmembers before spaceflight and for 6 mo following a 520-day spaceflight mission (74). The study reported that only two of the crewmembers returned to original microbiome configurations after landing (74). Altogether, these studies seem to suggest that shorter duration spaceflight may be associated with transient microbiota changes, whereas longer-duration spaceflight may be associated with more permanent changes in microbiome composition. However, the clinical significance of such changes remains elusive.

MICROGRAVITY-RELATED CARDIAC ARRHYTHMIA

Arrhythmias such as atrial fibrillation and ventricular tachycardia (VT) have been reported during spaceflight, perhaps secondary to hypokalemia, microgravity, autonomic nervous system changes, physical stress, and radiation during spaceflight (75). For instance, bigeminal premature ventricular contractions and premature atrial contractions were reported during the Apollo 15 mission, although it is important to note the lack of potassium consumption during this space mission (76, 77).

During spaceflight, QT prolongation can predispose to VT. QT prolongation indicates issues with ventricular repolarization perhaps secondary to bradycardia and QT-prolonging medications such as fluroquinolones, haloperidol, and sertraline (75). Indeed, microwave T-wave alternans was reported to increase during 2-wk head-down bed rest (78). Another study of 42 participants subjected to head-down bed rest for 60 days reported oscillations of ventricular repolarization (79). On the other hand, a study of 59 astronauts during the MIR program over 6 mo reported no electrocardiographic changes (80). Respress et al. (49) studied long-term microgravity exposure in mice via HU and reported an increase in pacing-induced nonsustained VT, with a trend toward QT prolongation, after 28 (30% incidence) and 56 (36% incidence) days of HU compared with the sham group (13% incidence).

Atrial fibrillation occurs in 5% of astronauts, which is a similar prevalence to the general population. A study of 13 astronauts conducted MRIs before and after 6 mo of spaceflight, with Holter monitoring over several 48-h time periods during spaceflight (35). Left atrial size transiently increased during spaceflight, but there were no effects on left atrial function. No astronauts developed atrial fibrillation, and there were no reported changes in P-wave duration or atrial ectopy.

One hypothesis by which spaceflight predisposes to cardiac arrhythmia is through accumulation of myocardial edema. While no study has directly looked at this yet, a prior study of two astronauts after 10-day spaceflight reported no changes in pulmonary tissue volume capillary blood flow from the C2H2-rebreathing test, indicating minimal accumulation of pulmonary interstitial edema (81). While this does not rule out the presence of cardiac interstitial edema, one may surmise based on the close vasculature connections between the heart and lungs that cardiac interstitial edema may be minimal during spaceflight. However, no study has directly shown this yet to our knowledge. Nonetheless, cardiac interstitial edema can predispose to arrhythmia development, and the presence of cardiac interstitial edema has been shown in dogs to induce collagen deposition (82), forming a reentry-prone substrate. Myocardial edema has also been shown to result in cardiac dysfunction that persists after resolution of the edema (83), which can predispose to arrhythmia secondary to myocardial ischemia and alterations in extracellular electrolyte and catecholamine levels.

Microgravity-Related Autonomic Changes

Autonomic dysfunction during and after microgravity exposure has been well reported in the literature (84). A study of four male subjects after 16 days of spaceflight reported decreased heart rate during spaceflight but increased heart rate persisting for 15 days after landing, indicative of sympathetic nervous system activation (85). During spaceflight, the ratio of low-frequency to high-frequency components of heart rate variability was similar to supine values, which was significantly lower than the ratio while standing, indicating parasympathetic control of heart rate during spaceflight. Moreover, the absence of postural changes under microgravity likely results in few instances of sympathetic activation and subsequently increases noradrenergic receptor sensitivity, thus accounting for the tachycardia observed upon landing.

In-flight measurements in astronauts taken at days 12 and 13 of spaceflight showed greater peroneal nerve muscle sympathetic activity, as well as plasma norepinephrine spillover (86). Another study reported a sharp increase in plasma and urinary catecholamines upon return from spaceflight, peaking at 8 days postlanding, despite no changes during spaceflight (87). However, a study of eight male astronauts reported no changes in plasma norepinephrine concentrations during and after ISS spaceflight missions, despite greater cardiac output and decreased blood pressure during spaceflight (2). Orthostatic intolerance after spaceflight was reported in a study of 40 astronauts after 16-day spaceflight, which demonstrated that astronauts unable to stand upright after landing

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exhibited the greatest venous norepinephrine levels (88). One explanation for microgravity-related orthostatic intolerance is decreased baroreflex-mediated chronotropic responses, which have been reported during and after short-term spaceflight (89). A study of 13 astronauts demonstrated a reduction in vagal baroreflex sensitivity assessed by R-R intervals changes on electrocardiogram in response to graded neck pressure and suction (89). Altogether, sympathetic activity seems to be increased in response to microgravity exposure and likely contributes to postspaceflight orthostatic intolerance.

CELLULAR MECHANISMS UNDERLYING MICROGRAVITY-INDUCED CARDIAC DYSFUNCTION

Cardiac Sarcomere and Contractile Proteins

The sarcomere is the basic contractile unit of cardiac muscle and is made up of a complex assembly of myofilament proteins. The interaction between thick myosin and thin actin filaments is the key force-generating process and is regulated by intracellular Ca^{2+} via the troponin-tropomyosin system (90). Sarcomeres are altered in many cardiac diseases such as hypertrophic cardiomyopathy, hypertension, heart failure, and cardiac atrophy (91). Regarding microgravity exposure, a 10-day spaceflight study of rats reported patchy loss of actin and myosin filaments and hypercontracted myofibrils after spaceflight (92), suggesting that microgravity may alter cardiac sarcomere structure.

Decreased myosin heavy chain and other myosin isoforms after microgravity conditions are expected based on what is previously known about cardiac atrophy (93). One of the initial studies of microgravity-related effects on the cardiac contractile apparatus was conducted in a 4-wk tail-suspension rat model (94). After tail suspension, rats exhibited unchanged myosin heavy chain, tropomyosin, tropomyosin binding protein, troponin T, and inhibitory troponin I despite decreased contractility and Ca^{2+}-activated ATPase activity (94). However, while no proteolytic fragments of tropomyosin and troponin T were detected in the tail-suspension group, there was increased NH_2-terminal truncated troponin I isoform (94), which lacks two protein kinase A (PKA) phosphorylation sites (serine-23 and serine-24). This observation may represent a functional adaptation of cardiac muscle in simulated microgravity that occurs via proteolytic regulation of cardiac myofibrillar proteins. Nonetheless, the results of this study (94) differed from other studies of cardiac atrophic remodeling, in which myosin heavy chain and the troponin-tropomyosin system are altered (66). Whether or not cardiac atrophy secondary to microgravity has differential effects on the cardiac sarcomere compared with cardiac atrophy secondary to other pathological states such as prolonged immobilization or deconditioning warrants further investigation.

Data from spaceflights also demonstrate a lack of alterations to thick myosin filaments in cardiac muscle after microgravity exposure (Fig. 1*D*) (94). Moreover, cardiac myosin isoform composition after 12 days (95) and 30 days (62) of spaceflight in gerbils demonstrated no significant differences relative to controls. There were also no differences in total myosin light chain and myosin binding protein content after 12 days in space (95).

To our knowledge, no studies to date have reported changes in thin actin filaments within cardiac muscle. Most of the research about actin has been focused on the β- and γ-actin isoforms, which are the nonmuscle isoforms that comprise part of the cytoskeleton (96, 97).

Titin, also known as connectin, is the third most abundant protein in muscle after myosin and actin and is a large molecule the size of half a sarcomere (3.0–3.7 MDa) (98). Titin contributes to muscular elasticity by modulating the operating range of sarcomere lengths and tension-related biochemical processes (98, 99). Titin is a key component in the assembly and functioning of vertebral striated muscle and plays a role in muscle development, signaling, and activity regulation (98).

Evidence from spaceflight experiments has demonstrated microgravity effects on titin filaments in skeletal (100) and cardiac (62, 101) muscle. In a 12-day gerbil spaceflight experiment, the ratio of long N_2BA to short N2B titin isoforms in the LV increased twofold, and titin phosphorylation was increased 1.3-fold (Fig. 1*D*) (101). Greater N_2BA isoform within the sarcomere I disk is known to increase cardiac muscle elasticity and thus contractile force according to the Franck–Starling law (102). Such microgravity-related titin effects could be compensatory to account for greater contractility required to pump the more viscous protein-rich blood during spaceflight (101). These positive inotropic effects on titin, however, may be countered by the increase in titin phosphorylation (101), which decreases the potentiating effect of titin on the actin-myosin ATPase (103). Indeed, titin-isolated gerbils after 12 days of spaceflight exhibited a lower activating effect on the actin-myosin ATPase, which was likely due to a combination of changes in secondary titin structure, as well as titin phosphorylation (101).

In contrast with these findings, a 30-day study of spaceflight in mice reported no differences in cardiac muscle titin protein expression and phosphorylation between the control and spaceflight groups (62). However, there was a twofold increase in titin gene expression in cardiac muscle (62), suggestive of greater titin turnover and/or degradation. The proportion of the cardiac N_2BA titin isoform also trended higher in the flight group, but the difference did not reach statistical significance (62). Altogether, it is important to note that the effect microgravity on cardiac muscle titin (62, 101) is less clear than that for skeletal muscle titin, which is decreased in content along with a shift from "slow" to "fast" myosin isoforms under microgravity conditions (62, 100).

Microgravity-Related Changes in Cardiac Calcium Handling

Ca^{2+} homeostasis is disrupted under states of microgravity: urinary Ca^{2+} excretion increases, intestinal Ca^{2+} absorption decreases, and serum Ca^{2+} increases (104). These changes occur in the absence of changes in parathyroid hormone and vitamin D metabolism and can affect the cardiac function because of its dependence on extracellular Ca^{2+} (105).

Normal Excitation-Contraction Coupling

Excitation-contraction coupling is the physiological mechanism by which electrical excitation of cardiac myocytes leads

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to muscle contraction (106, 107). The Ca^{2+} release unit forms the basic functional unit of excitation-contraction coupling occurs and refers to specialized Ca^{2+}-channel containing areas within the cardiomyocyte sarcolemmal membrane harboring transverse tubules containing L-type Ca^{2+} channels (LTCCs), sarcoplasmic reticulum (SR)-containing ryanodine receptor-2 (RyR2), and sarco(endo)plasmic reticulum Ca^{2+}-ATPase-2a (SERCA2a), among other components (108). Ca^{2+}-induced Ca^{2+} release (CICR) is a key distinction between cardiac and skeletal muscle, in which LTCCs and RyRs are mechanically linked such that Ca^{2+} influx via LTCC is not necessary to trigger SR Ca^{2+} release. In contrast to skeletal myocytes, cardiomyocytes rely on Ca^{2+} entry through LTCCs to trigger RyR2 opening and subsequent release of roughly 50% of SR Ca^{2+} content (109), resulting in the release of troponin I from the myofilament and actin-myosin cross-bridge cycling with subsequent myofilament contraction.

Microgravity-Related Effects on Ca^{2+} Release

Ca^{2+} influx via the LTCC is stimulated by β-adrenergic-mediated PKA phosphorylation (110) and decreased by protein kinase C (PKC) phosphorylation (111). Alterations in LTCC function seem to contribute to microgravity-related cardiac Ca^{2+} mishandling (49, 55, 112). While 2 wk of weightlessness via head-down tilt was reported to increase the chronotropic and vasodilatory response to isoproterenol in humans (58), a 4-wk tail-suspension study in rats demonstrated a blunted inotropic response to isoproterenol (113), indicating the importance of the length of microgravity exposure, as well as differential chronotropic versus inotropic effects of microgravity. A 4-wk tail-suspension experiment in rats also reported attenuated LTCC Ca^{2+} transients (112) and cAMP production (55) in response to isoproterenol (Fig. 2). Interestingly, PKA expression and membrane translocation in response to isoproterenol were unaltered in a 4-wk tail-suspension study in rats (113) despite the reduction in intracellular peak Ca^{2+} transient. Rather than changes in PKA activity and/or expression, depressed cardiac function in the tail-suspension group was hypothesized to be secondary to cardiac troponin NH_2-terminal degradation (113). A more recent study further attributed decreased myocardial nitric oxide synthase expression and activity as a mechanistic driver of microgravity-related attenuation in LTCC function (Fig. 2) (114). Decreased nitric oxide synthase reduced nitric oxide-mediated *S*-nitrosylation of the LTCC a_1-subunit, which protects the LTCC against oxidative damage and changes in gating properties under physiological conditions (114).

RyR2 is pivotal for systolic Ca^{2+} release from the SR and forms the center of a large macromolecular channel complex comprised of several intra-SR and cytosolic regulatory binding partners (115, 116). Like LTCC, alterations in RyR2 function seem to contribute to microgravity-related cardiac Ca^{2+} mishandling (49, 55, 112). A study aimed at exploring the role of microgravity on smooth muscle cells examined rats after 8 days in the ISS (117). Myocytes cultured from the hepatic portal vein of these mice had decreased RyR1 expression, with concomitant attenuation of CICR, phenotypically mimicking the effects of smooth muscle cells treated with antisense oligonucleotides against RyR1 (105). Consistent with this finding, spontaneously hypertensive rats exhibited the opposite findings, upregulated RyR1 with concomitantly increased CICR (117). Altogether, these results indicate that smooth muscle cells adapt Ca^{2+} signaling to microgravity conditions by modulating RyR1 expression, specifically decreasing RyR1, CICR, and thus contractility in response to microgravity.

A study on microgravity effects on the main cardiac RyR2 isoform within cardiomyocytes was conducted using tail suspension for 4 to 8 wk in mice (49). Tail-suspended mice exhibited prolonged QT intervals and premature ventricular contractions (Fig. 2) on electrocardiogram but no spontaneous

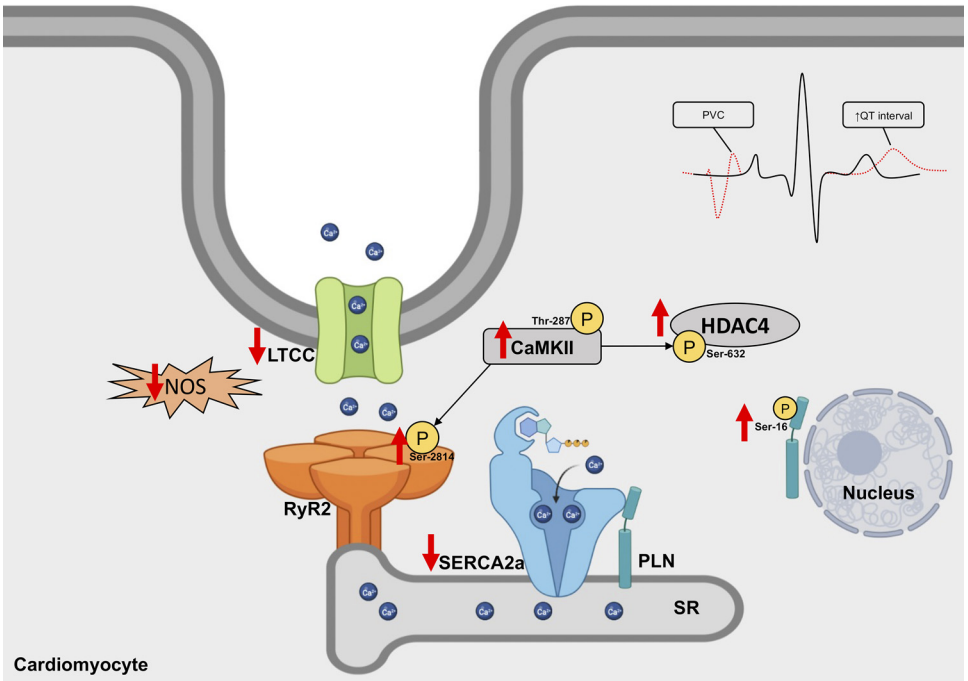

Figure 2. Microgravity-related changes within the cardiomyocyte. Microgravity leads to attenuated cardiac function because of decreased nitric oxide synthase (NOS) activity, predisposing L-type calcium channels (LTCCs) to oxidative damage, and ryanodine receptor 2 (RyR2) Ca^{2+} leak, secondary to increased calcium/calmodulin-dependent protein kinase II (CaMKII) phosphorylation at serine-2814. Increased phosphorylation of CaMKII at threonine-287 and histone deacetylase 4 (HDAC4) at serine-632 results in greater RyR2 activity and cardiac remodeling, respectively. Decreased sarco(endo)plasmic reticulum calcium ATPase 2a (SERCA2a) activity results in decreased sarcoplasmic reticulum (SR) Ca^{2+} content. These changes contribute to the presence of premature ventricular contraction (PVCs) and prolonged QT interval on electrocardiogram (ECG).

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arrhythmias. Tail-suspended mice also had a greater frequency of spontaneous ventricular SR Ca^{2+} release events and SR Ca^{2+} leak. RyR2 phosphorylation at serine-2814 and CaMKII autophosphorylation at threonine-287 were both increased in the HU group, suggesting increased RyR2 activity (Fig. 2). Taken together, these findings indicate that CaMKII-mediated phosphorylation of RyR2 contributes to arrhythmia susceptibility via RyR2 Ca^{2+} leak following microgravity exposure.

Microgravity-Related Changes in Ca^{2+} Reuptake

In addition to RyR2 and LTCC Ca^{2+} handling, diastolic Ca^{2+} handling is disrupted under states of microgravity (118, 119). Briefly, diastolic Ca^{2+} handling is regulated by SERCA2a and its inhibitor, phospholamban (PLN) (120). Phosphorylation of PLN serine-16 or threonine-17 by PKA secondary to β-adrenergic signaling relieves PLN inhibition of SERCA2a, increasing diastolic SR Ca^{2+} reuptake and increasing RyR2-mediated systolic SR Ca^{2+} release (120, 121).

A prior study of 4-wk tail suspension in rats provided a link between intracellular Ca^{2+} regulation and cardiomyocyte apoptosis during recovery following microgravity exposure (118). Upon exposure to normal gravity after microgravity, there was a catecholamine surge to compensate for the reduced blood volume secondary to microgravity-induced cephalic blood redistribution. This surge was a key mediator of cardiomyocyte apoptosis due to increased intracellular Ca^{2+} transients and subsequent transcriptional upregulation of apoptosis-related genes such as calpain-2 (118). The study also reported that isoproterenol increased serine-16-phosphorylated PLN localization at the nuclear envelope in the tail-suspension group (118). This led to an increase in nuclear Ca^{2+} transient amplitude in response to isoproterenol. Intranuclear Ca^{2+} is a key regulator of cardiac gene regulation, and perhaps expression of alternative RyR2 splice variants helps explain some of the microgravity-related cardiac functional and apoptotic changes (122).

A more recent study on HL-1 cardiomyocytes used a clinostat to study microgravity effects on Ca^{2+} handling. HL-1 cells exposed to microgravity were smaller, indicative of cardiomyocyte atrophy, and exhibited more spontaneous cytosolic Ca^{2+} oscillations, indicative of diastolic Ca^{2+} mishandling (123). Indeed, phosphorylation of CaMKII at threonine-287 was greater in HL-1 cells exposed to microgravity, both of which predispose to spontaneous Ca^{2+} sparks (124), as well as histone deacetylase 4 (HDAC4) phosphorylation at serine-632 (Fig. 2) (125). HDAC4 phosphorylation resulted in transcriptional upregulation of cardiac remodeling genes including atrial natriuretic peptide and B-type natriuretic peptide (126, 127).

A 2-wk rat-tail suspension study similarly reported greater Ca^{2+} transient amplitude despite unchanged cardiac sarcoplasmic reticulum Ca^{2+} content (128). This increase in Ca^{2+} transient amplitude was shown to be due to greater RyR2 phosphorylation and CICR (128). Interestingly, a more recent 4-wk tail-suspension experiment in mice reported that SR Ca^{2+} trended lower in the microgravity group, but the results were not statistically significant (49). A reduced SR Ca^{2+} content indicates diastolic SR Ca^{2+} leak, perhaps secondary to CaMKII phosphorylation of RyR2 at serine-2814 as previously reported (129). Diastolic Ca^{2+} leak was also reportedly caused by decreased SERCA2a activity in a 4-wk rat-tail suspension study (Fig. 2) (130). Taken together, decreases in SR Ca^{2+} content seem to parallel the degree of cardiac atrophy with increasing microgravity exposure duration (29).

To study microgravity effects at a cellular level, Wnorowski et al. (66) housed hiPSC-CMs for 1 mo on the ISS. The microgravity-exposed cells exhibited no changes in sarcomeric structure, RyR2 and SERCA2a expression, or Ca^{2+} transient amplitude. However, the microgravity-exposed hiPSC-CMs exhibited prolongation of Ca^{2+} transient decay, as well as heartbeat irregularity, indicative of diastolic Ca^{2+} mishandling perhaps secondary to RyR2 leak (66). Troponins T and I1 were upregulated in flight, but other sarcomeric genes such as *MYH7* were expressed at a similar rate to ground controls. Motif enrichment analyses revealed upregulations in myocyte enhancer factor 2 and specificity protein 1 motifs in the microgravity-exposed hiPSC-CMs, both contributing to a hypertrophic phenotype (66).

Altogether, it seems likely that abnormal cardiomyocyte intracellular Ca^{2+} handling occurs because of increased CaMKII phosphorylation of RyR2 and decreased SERCA2a activity, predisposing to cardiac dysfunction and arrhythmia following microgravity conditions.

Conclusions

In this review, we have summarized key findings from studies demonstrating the effects of microgravity on cardiac structure, function, and cardiomyocyte Ca^{2+} handling. Our review provides a framework for future spaceflight studies, and the results we summarize in this review can be further applied to nonspaceflight conditions such as prolonged bed rest and immobilization.

DATA AVAILABILITY

Data sharing will be made available upon request from the authors.

GRANTS

This work was supported by National Heart, Lung, and Blood Institute Grants R01-HL147108, R01-HL153350, and R01-HL089598 (to X.H.T.W.); the Robert and Janice McNair Foundation McNair MD/PhD Scholars Program (to J.A.K.); and the Baylor College of Medicine Medical Scientist Training Program (to J.A.K.).

DISCLOSURES

X.H.T.W. is a founding partner of Elex Biotech, a start-up company that developed drug molecules to target ryanodine receptors to treat cardiac arrhythmias. The remaining authors have nothing to disclose.

AUTHOR CONTRIBUTIONS

X.H.T.W. conceived and designed research; J.A.K. prepared figures; M.R.S. and J.A.K. drafted manuscript; J.P.S. and X.H.W. edited and revised manuscript; M.R.S., J.A.K., J.P.S., and X.H.T.W. approved final version of manuscript.

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